Two-photon transitions and more ..

Bloch equation

$$\frac{d\vec{M}^r}{dt} = \vec{M}^r \times (\gamma_e \vec{B}^r - \vec{\omega}_r),$$

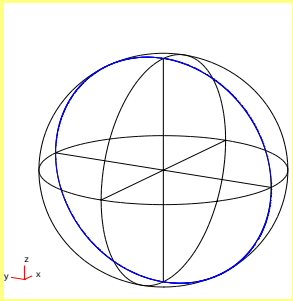
where $\vec{B}^r = \hat{R}(\vec{B}_0 + \vec{B}_1(t))$ and $\vec{\omega}_r$ is the angular velocity of the rotating frame.

Conventional resonance

$\omega_r = \omega_0 = \gamma_e B_0$,

$\vec{\omega}_1^r = \gamma_e B_1 [1, 0, 0]$,

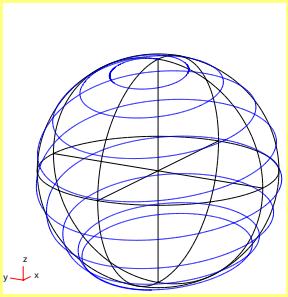
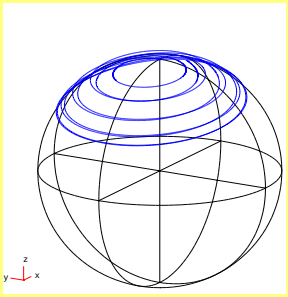
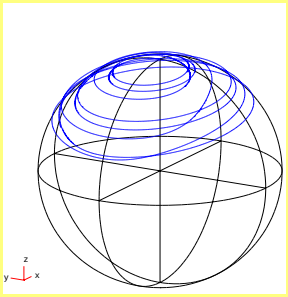
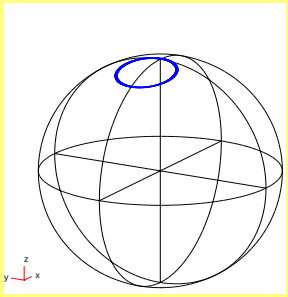
$\omega_1^r t_p = \gamma_e B_1 t_p = 2\pi.$



Two-photon (σ^+-π) transitions

$\omega_r = \omega_{mw}$, $\omega_{mw} - \omega_0 = 8\omega_1^r$, $\omega_1^r t_p = 2\pi.$

$\vec{\omega}_{2p}^r = 2\omega_1^r [1, 0, 2\cos(8\omega_1^r t)]$ $[1, 0, 2\cos(9\omega_1^r t)]$ $[1, 0, 2\cos(7\omega_1^r t)]$ $[1, 0, 0]$

Three-photon (σ^+–σ^-–σ^+) resonances

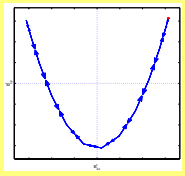
4-color irradiation (G. Goelman 87)

$\omega_r = \omega_0 = center$

$\vec{\omega}_{3p}^r = 2\omega_1^r [\cos(8\omega_1^r t + \alpha), \cos(4\omega_1^r t + \beta), 0]$

$\omega_1^r t_p = 8\pi$

$\alpha = 0$
$\beta = 0$



$\alpha = \pi/2$
$\beta = 0$

